Dr. Kevin K. Hardegree-Ullman

Exoplanets Stars Brown Dwarfs

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Education

Ph.D. Physics, University of Toledo 2018
Dissertation: *Kepler Planet Occurrence Rates for Mid-Type M Dwarfs as a Function of Spectral Type*
Advisor: Prof. Michael C. Cushing

B.S. Astronomy, University of Arizona 2011

Professional History

Assistant Research Scientist, Caltech/IPAC-NASA Exoplanet Science Institute 2024–Present

Postdoctoral Research Associate, University of Arizona/Steward Observatory 2021–2024
Advisor: Prof. Dániel Apai

Postdoctoral Scholar, Caltech/IPAC-NExScI 2018–2021
Advisor: Dr. Jessie L. Christiansen

Visiting Graduate Student Research Fellow, IPAC 2017
Advisor: Dr. Jessie L. Christiansen

Ritter Observatory Research Assistant, University of Toledo 2013–2017
Observatory Director: Prof. Jon Bjorkman

Awards and Grants

US Contributions to Ariel Preparatory Science Grant – Science Lead 2025–2027
Research grant to characterize high-priority targets for the upcoming Ariel mission with high-resolution spectroscopy. **$121,598**

NASA WIYN PI Data Awards – 7 total awards 2015–2018
NASA/NSF Exoplanet Observational Research Program awards to characterize M dwarfs for exoplanet demographics studies using the WIYN telescope. **$109,400**

Telescope Time as PI

SOAR - NOIRLab Survey Program 2019–2022
Atmospheric survey of hot Jupiters observed by TESS (Goodman). 30 nights
3 Publications: #16, 17, 25

IRTF 2016–2026
M dwarfs and metallicity characterization projects (SpeX). 47 nights
3 Publications: #42, 45, 49

WIYN 2015–2018
Kepler M dwarfs and TESS cool star characterization projects (Hydra). 46 nights
1 Publication: #42

Lowell Discovery Telescope 2013–2017
Kepler M dwarfs and transiting exoplanet projects (DeVeny, LMI). 20 nights
2 Publications: #4, 42

Service

ExoPAG SIG 2 - Demographics Working Group	2025–Present
IPAC Student Coordinator	2025–Present
Las Cumbres Observatory Time Allocation Committee	2025-Present
SOAR Classification Spectrograph Working Group	2023–Present
NOIRLab Mid-Scale Observatories Users' Committee	2022–Present
SOAR Telescope Science Advisory Committee (***Chair***, 2025–Present)	2020–Present
Subject-matter expert in NASA peer reviews	2020–Present
ExoPAG SAG 22 - Existing Stellar Catalogs Task Force Co-Lead	2020–2021
Referee for AAS Journals, MNRAS, PASP, and A & A	2018–Present

Advising/Mentoring History

Undergraduate Advising

Vivian Welch, University of Arizona	2024
Bryan Wang, University of Arizona	2024
Justin Klingele, University of Arizona	2021–2023
Malhar Dave, University of Arizona	2022
Kate Wienke, Kansas University	2022
Jillian Henkel, University of Toronto	2022

Graduate Mentoring

Brandon Radzom, Ph.D. Candidate, Indiana University *–IPAC Visiting Graduate Student Research Fellow*	2026
Galen Bergsten, Ph.D., University of Arizona, LPL	2022–2025
Kiersten Boley, Ph.D., The University of Ohio	2021–2024
Rachel Fernandes, Ph.D., University of Arizona, LPL	2020–2023
Jon Zink, Ph.D., University of California, Los Angeles	2019–2021

Teaching

Summer Science Program - Guest Instructor (Exoplanets)	2023–2024
ASTR 399 Astronomy Independent Study - Research Instructor	2022–2023
PHYS 1910 Frontiers Of Physics And Astronomy - Guest Instructor (Exoplanets)	2013–2016
ASTR 5880 Astrophysical Measurements - Guest Instructor (3-color images)	2013
ASTR 2050 Elementary Astronomy Laboratory - Instructor	2013
ASTR 1010 Survey of Astronomy - Teaching Assistant	2012–2013
PHYS 2070 General Physics Laboratory - Instructor	2012

Publication Statistics

59 Papers | 5 Research Notes | 1 Report to NASA | 2153 Citations | Link to ADS

Major Contributions

65. *COUNTESS. I. A Uniformly Vetted Catalog of Known and New Transiting Exoplanets in the TESS Northern Continuous Viewing Zone*
Hotnisky, A.; Fernandes, R. B.; **Hardegree-Ullman, K. K.**; et al. 2026. ApJS, 286, 42

64. *Exoplanets in ancient stellar populations: occurrence constraints and hot-Jupiter candidates in the Galactic halo*
Bashi, D.; Kunimoto, M.; **Hardegree-Ullman, K. K.**; et al. 2026. MNRAS, 551, 1386

63. *A Spitzer Space Telescope Exploration Science Program to Search for Y-dwarf Variability*
Cushing, M. C.; Trucks, J. L.; **Hardegree-Ullman, K. K.**; et al. 2026. AJ, 172, 166

62. *Gaia DR3 IDs for TESS Input Catalog Targets*
Hardegree-Ullman, K. K.; Swain, M.; Christiansen, J. L. et al. 2026. RNAAS, 10, 77

61. *Scaling K2 VIII: Short-Period Sub-Neptune Occurrence Rates Peak Around Early-Type M Dwarfs*
Hardegree-Ullman, K. K.; Bergsten, G. J.; Christiansen, J. L. et al. 2025. AJ, 170, 183

60. *Bioverse: Giant Magellan Telescope and Extremely Large Telescope Direct Imaging and High-resolution Spectroscopy Assessment—Surveying Exo-Earth O_2 and Testing the Habitable Zone Oxygen Hypothesis*
Hardegree-Ullman, K. K.; Apai, D.; Haffert, S. Y. et al. 2025. AJ, 169, 171

59. *The First Evidence of a Host Star Metallicity Cutoff in the Formation of Super-Earth Planets*
Boley, K. M.; Christiansen, J. L.; Zink, J. K.; **Hardegree-Ullman, K. K.** et al. 2024. AJ, 168, 128

58. *Scaling K2. VII. Evidence For a High Occurrence Rate of Hot Sub-Neptunes at Intermediate Ages*
Christiansen, J. L.; Zink, J. K.; **Hardegree-Ullman, K. K.** et al. 2023. AJ, 166, 248

57. *No Evidence for More Earth-sized Planets in the Habitable Zone of Kepler's M versus FGK Stars*
Bergsten, G. J.; Pascucci, I.; **Hardegree-Ullman, K. K.** et al. 2023. AJ, 166, 234

56. *Using Photometrically-Derived Properties of Young Stars to Refine TESS's Transiting Young Planet Survey Completeness*
(Fernandes, R. B. & **Hardegree-Ullman, K. K.** co-first authors); Pascucci, I. et al. 2023. AJ, 166, 175

55. *Bioverse: A Comprehensive Assessment of the Capabilities of Extremely Large Telescopes to Probe Earth-like O_2 Levels in Nearby Transiting Habitable Zone Exoplanets*
Hardegree-Ullman, K. K.; Apai, D.; Bergsten, G. J. et al. 2023. AJ, 165, 267

54. *Scaling K2. VI. Reduced Small-planet Occurrence in High-galactic-amplitude Stars*
Zink, J. K.; **Hardegree-Ullman, K. K.**; Christiansen, J. L. et al. 2023. AJ, 165, 262

53. *EDEN Survey: Small Transiting Planet Detection Limits and Constraints on the Occurrence Rates for Late M Dwarfs within 15 pc*
Dietrich, J.; Apai, D.; Schlecker, M.; **Hardegree-Ullman, K. K.** et al. 2023. AJ, 165, 149

52. *Scaling K2. V. Statistical Validation of 60 New Exoplanets From K2 Campaigns 2-18*
Christiansen, J. L.; Bhure, S.; Zink, J. K.; **Hardegree-Ullman, K. K.** et al. 2022. AJ, 163, 244

51. *Final Report for SAG 22: A Target Star Archive for Exoplanet Science*
Hinkel, N. R.; Pepper, J.; Stark, C. C. et al. 2021. Final report to guide NASA exoplanet science.

50. *Scaling K2. IV. A Uniform Planet Sample for Campaigns 1-8 and 10-18*
Zink, J. K.; **Hardegree-Ullman, K. K.**; Christiansen, J. L. et al. 2021. AJ, 162, 259

49. *K2-138 g: Spitzer Spots a Sixth Planet for the Citizen Science System*
Hardegree-Ullman, K. K.; Christiansen, J. L.; Ciardi, D. R. et al. 2021, AJ, 161, 219

48. *Scaling K2. III. Comparable Planet Occurrence in the FGK Samples of Campaign 5 and Kepler*
Zink, J. K.; **Hardegree-Ullman, K. K.**; Christiansen, J. L. et al. 2020, AJ, 160, 94

47. *Scaling K2. II. Assembly of a Fully Automated C5 Planet Candidate Catalog Using EDI-Vetter*
Zink, J. K.; **Hardegree-Ullman, K. K.**; Christiansen, J. L. et al. 2020, AJ, 159, 154

46. *Scaling K2. I. Revised Parameters for 222,088 K2 Stars and a K2 Planet Radius Valley at 1.9 $R_{\oplus}$*
Hardegree-Ullman, K. K.; Zink, J. K.; Christiansen, J. L. et al. 2020, ApJS, 247, 28

45. *Characterizing K2 Candidate Planetary Systems Orbiting Low-mass Stars. IV. Updated Properties for 86 Cool Dwarfs Observed during Campaigns 1–17*
Dressing, C. D.; **Hardegree-Ullman, K. K.**; Schlieder, J. E. et al. 2019, AJ, 158, 87

44. *Bright Opportunities for Atmospheric Characterization of Small Planets: Masses and Radii of K2-3 b, c, and d and GJ3470 b from Radial Velocity Measurements and Spitzer Transits*
Kosiarek, M. R.; Crossfield, I. J. M.; **Hardegree-Ullman, K. K.** et al. 2019, AJ, 157, 97

43. *Catalog of New K2 Exoplanet Candidates from Citizen Scientists*
Zink, J. K.; **Hardegree-Ullman, K. K.**; Christiansen, J. L. et al. 2019, RNAAS, 3, 43

42. *Kepler Planet Occurrence Rates for Mid-type M Dwarfs as a Function of Spectral Type*
Hardegree-Ullman, K. K.; Cushing, M. C.; Muirhead, P. S.; Christiansen, J. L. 2019, AJ, 158, 75

41. *Spitzer Light Curves of the Young, Planetary-Mass TW Hya Members 2MASS J11193254-1137466AB and WISEA J114724.10-204021.3*
Schneider, A. C.; **Hardegree-Ullman, K. K.**; Cushing, M. C. et al. 2018, ApJ, 155, 238

40. *Investigating the physical properties of transiting hot Jupiters with the 1.5-m Kuiper Telescope*
Turner, J. D.; Leiter, R. M.; Biddle, L. I. et al. 2017, MNRAS, 472, 3871

39. *Observed Variability at 1μm and 4μm in the Y0 Brown Dwarf WISEP J1738+2732*
Leggett, S. K.; Cushing, M. C.; **Hardegree-Ullman, K. K.** et al. 2016, ApJ, 830, 141

38. *Ground-Based Near-UV Observations of 15 Transiting Exoplanets: Constraints on Their Atmospheres and No Evidence for Asymmetrical Transits*
Turner, J. D.; Pearson, K. A.; Biddle, L. I. et al. 2016, MNRAS, 459, 789

37. *The First Detection of Photometric Variability in a Y Dwarf: WISE J1405+5534*
Cushing, M. C.; **Hardegree-Ullman, K. K.**; Trucks, J. L. et al. 2016, ApJ, 823, 152

36. *Near-UV and Optical Observations of the Transiting Exoplanet TrES-3b*
Turner, J. D.; Smart, B. M.; **Hardegree-Ullman, K. K.** et al. 2013, MNRAS, 428, 678

35. *A New Online Astronomy Resource for Education and Outreach*
Impey C. D.; **Hardegree-Ullman, K. K.**; Patikkal, A. et al. 2013, Astro. Ed. Rev., 12, 010301

Minor Contributions

34. *Modeling the Impact of Unresolved Stellar Companions on Detection Sensitivity in Kepler's Small-planet Occurrence Rates*
Bergsten, G. J.; Ciardi, D. R.; Christiansen, J. L. et al. 2026. AJ, 171, 110

33. *No Turnover Found in Giant-Planet Occurrence Within 3 au Around Low-mass Stars*
Gonglewski, K. L.; Bergsten, G. J.; Pascucci, I. et al. 2025. RNAAS, 9, 356

32. *Bioverse: Assessing the Ability of Direct Imaging Surveys to Empirically Constrain the Habitable Zone via Trends in Albedo*
Tuchow, N. W.; Stark, C. C.; Apai, D. et al. 2025. AJ, 170, 271

31. *Bioverse: Potentially Observable Exoplanet Biosignature Patterns under the UV Threshold Hypothesis for the Origin of Life*
Schlecker, M.; Apai, D.; Affholder, A. et al. 2025. ApJ, 987, 24

30. *The NASA Exoplanet Archive and Exoplanet Follow-up Observing Program: Data, Tools, and Usage*
Christiansen, J. L.; McElroy, D. L.; Harbut, M. et al. 2025. PSJ, 6,186

29. *Concerning the possible exomoons around Kepler-1625 b and Kepler-1708 b*
Kipping, D.; Teachey, A..; Yahalomi, D. A. et al. 2025. Nature Astronomy, 9, 795

28. *GEMS JWST: Transmission spectroscopy of TOI-5205b reveals significant stellar contamination and a metal-poor atmosphere*
Canas, C. I.; Lustig-Yaeger, J.; Tsai, S-M et al. 2025. AJ, submitted

27. *Planet Masses, Radii, and Orbits from NASA's K2 Mission*
Howard, A. W.; Sinukoff, E.; Blunt, S. et al. 2025. ApJS, 278, 52

26. *Signatures of Atmospheric Mass Loss and Planet Migration in the Time Evolution of Short-Period Transiting Exoplanets*
Fernandes, R. B.; Bergsten, G. J.; Mulders, G. D. et al. 2025. AJ, 169, 208

25. *The Palomar twilight survey of 'Ayló'chaxnim, Atiras, and comets*
Bolin, B. T.; Masci, F. J.; Coughlin, M. W. et al. 2025. Icarus 425, 116333

24. *New ephemerides and detection of transit-timing variations in the K2-138 system using high-precision CHEOPS photometry*
Vivien, H. G.; Hoyer, S.; Deleuil, M. et al. 2024. A & A 668, 192

23. *Metallicities and Refined Stellar Parameters for 52 Cool Dwarfs with Transiting Planets and Planet Candidates*
Gore, R.; Giacalone, S.; Dressing, C. D. et al. 2024. ApJS 271, 48

22. *A Reply to: Large Exomoons unlikely around Kepler-1625 b and Kepler-1708 b*
Kipping, D.; Teachey, A.; Yahalomi, D. A. et al. 2024. Submitted to Nature Astronomy.

21. *Clouds and Clarity: Revisiting Atmospheric Feature Trends in Neptune-size Exoplanets*
Brande, J.; Crossfield, I. J. M.; Kreidberg, L. et al. 2023. ApJL, 961, 23

20. *Bioverse: The Habitable Zone Inner Edge Discontinuity as an Imprint of Runaway Greenhouse Climates on Exoplanet Demographics*
Schlecker, M.; Apai, D.; Lichtenberg, T. et al. 2023. PSJ, 5, 3

19. *Water Absorption in the Transmission Spectrum of the Water World Candidate GJ 9827 d*
Roy, P.-A.; Benneke, B.; Piaulet, C. et al. 2023. ApJ, 954, 52

18. *The mass of TOI-519 b: A close-in giant planet transiting a metal-rich mid-M dwarf*
Kagetani, T.; Narita, N.; Kimura, T. et al. 2023. PASJ, 75, 713

17. *Preliminary estimates of the Zwicky Transient Facility 'Ayló'chaxnim asteroid population completeness*
Bolin, B. T.; Ahumada, T.; van Dokkum, P. et al. 2023. Icarus, 394, 115442

16. *The discovery and characterization of (594913) 'Ayló'chaxnim, a kilometre sized asteroid inside the orbit of Venus*
Bolin, B. T.; Ahumada, T.; van Dokkum, P. et al. 2022. MNRAS, 517, 49

15. *The K2-3 System Revisited: Testing Photoevaporation and Core-powered Mass Loss with Three Small Planets Spanning the Radius Valley*
Diamond-Lowe, H.; Kreidberg, L.; Harman, C. E. et al. 2022. AJ, 164, 172

14. *Tentative Evidence for Water Vapor in the Atmosphere of the Neptune-Size Exoplanet HD 106315 c*
Kreidberg, L.; Mollière, P.; Crossfield, I. J. M. et al. 2022. AJ, 164, 124

13. *pterodactyls: A Tool to Uniformly Search and Vet for Young Transiting Planets In TESS Primary Mission Photometry*
Fernandes, R. B.; Mulders, G. D.; Pascucci, I. et al. 2022. AJ, 164, 78

12. *An Updated Ephemeris for K2-138 d*
Boyle, A.; Christiansen, J. L.; Vissapragada, S.; **Hardegree-Ullman, K. K.** 2022. RNAAS, 6, 71

11. *An exomoon survey of 70 cool giant exoplanets and the new candidate Kepler-1708 b-i*
Kipping, D.; Bryson, S.; Burke, C. et al. 2022. Nature Astronomy, 6, 367

10. *WASP-107b's Density Is Even Lower: A Case Study for the Physics of Planetary Gas Envelope Accretion and Orbital Migration*
Piaulet, C.; Benneke, B.; Rubenzahl, R. A. et al. 2021, AJ, 161, 70

9. *Transmission Spectroscopy for the Warm Sub-Neptune HD 3167c: Evidence for Molecular Absorption and a Possible High Metallicity Atmosphere*
Evans, T. M.; Crossfield, I. J. M.; Benneke, B. et al. 2021, AJ, 161, 18

8. *Revisiting the HIP 41378 System with K2 and Spitzer*
Berardo, D.; Crossfield, I. J. M.; Werner, M. et al. 2019, AJ, 157, 185

7. *An Updated Ephemeris for the Single-lined Orbit of the Supergiant μ Sagittarii*
Johnson, R. A.; Richardson, N. D.; Moffat, A. F. J. et al. 2018. RNAAS, 2, 138

6. *Characterizing K2 Candidate Planetary Systems Orbiting Low-mass Stars. III. A High Mass and Low Envelope Fraction for the Warm Neptune K2-55b*
Dressing, C. D.; Sinukoff, E.; Fulton, B. J. et al. 2018, AJ, 156, 70

5. *Planetary Candidates from K2 Campaign 16*
Yu, L.; Crossfield, I. J. M.; Schlieder, J. E. et al. 2018, AJ, 156, 22

4. *2MASS J11151597+1937266: A Young, Dusty, Isolated, Planetary-mass Object with a Potential Wide Stellar Companion*
Theissen, C. A.; Burgasser, A. J.; Bardalez Gagliuffi, D. C. et al. 2018, ApJ, 853, 75

3. *The most massive heartbeat: an in-depth analysis of ι Orionis*
Pablo, H.; Richardson, N. D.; Fuller, J. et al. 2017, MNRAS, 467, 2494

2. *A Spectroscopic Orbit for the Late-type Be Star β CMi*
Dulaney, N. A.; Richardson, N. D.; Gerhartz, C. J. et al. 2017, ApJ, 836, 112

1. *Photometric Monitoring of the Coldest Known Brown Dwarf with the Spitzer Space Telescope*
Esplin, T. L.; Luhman, K. L.; Cushing, M. C. et al. 2016, ApJ, 832, 58